

RAHUL SARKAR

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IIT (ISM) Dhanbad

Education

Indian Institute of Technology (ISM) Dhanbad

Expected May 2027

Master of Technology in Fuel and Energy Engineering (GPA: 8.41 / 10.00)

Dhanbad, Jharkhand

- **Relevant Coursework:** Computational Fluid Dynamics, Alternate Energy Systems, Bio-Fuels, Fuel Technology, Processing of Fuels, Power Plant Engineering.
- **Focus Areas:** Sustainable fuels (SAF, Bio-LNG), Carbon Capture, Utilization & Storage (CCUS), Pyrolysis, and Renewable Energy.

Kalyani Government Engineering College

July 2020

Bachelor of Technology in Mechanical Engineering (GPA: 7.85 / 10.00)

Kalyani, West Bengal

- **Relevant Coursework:** Thermodynamics, Heat & Mass Transfer, Material Handling, CAD/CAM.

Technical Skills

Energy & Domain Expertise: Bio-fuels, CCUS, Renewable Energy, Pyrolysis, Carbon Credit, Computational Fluid Dynamics (FVM, FDM), Life Cycle Assessment

Languages & Scripting: Python (Pandas, NumPy, OpenCV, Selenium, PyPDF2), SQL

Data Engineering & Platforms: PySpark, Databricks, Informatica PowerCenter, AWS (Redshift, SageMaker, S3), Relational Databases

Analytics & Modeling: Machine Learning, Data Visualization (matplotlib), MS-Excel, MS-PowerPoint, Git

Soft Skills: Stakeholder Management, Technical Synthesis, Storytelling, Cross-Functional Collaboration, Client Handling

Experience

Tata Consultancy Services Limited

Jan 2021 – May 2025

Engineering Change and Data Analyst

Kolkata, West Bengal

- Developed robust data pipelines using Python, SQL, and Pandas to collect, clean, and validate massive supply chain datasets.
- Engineered Python and OCR-based automation tools to extract unstructured data, slashing manual workflows and boosting quality.
- Translated operational analytics into process optimizations, increasing shipment output by 150% via actionable data insights.
- Led cross-functional stakeholder collaboration and mentored associates to guarantee seamless execution of client deliverables.

Data Engineer

- Architected scalable ETL pipelines and structured databases in Databricks, centralizing external datasets into Unity Catalog.
- Migrated legacy Informatica PowerCenter workflows to Databricks, cutting infrastructure costs by 60% and accelerating processing.
- Deployed automated Python and PySpark cluster optimization scripts, drastically reducing compute energy usage and runtime.
- Integrated multi-source data via FiveTran, APIs, and S3 Autoloader to translate complex datasets into business intelligence.
- Streamlined CI/CD deployments using GitLab and authored comprehensive API documentation, dropping troubleshooting time by 40%.

Projects

Critical Mineral Mapping & Market Analysis Dashboard | Python, Streamlit, Pandas, Data Analysis

- Developed an interactive dashboard to visualize state-wise production and import dependency matrices for critical energy transition minerals.
- Conducted commodity market analysis by integrating trade partner networks and forecasting demand-supply gaps to derive actionable insights.

Migration of ETL Workflows & Scalable Data Pipelines | Databricks, PySpark, Python, SQL, FiveTran

- Architected the migration of legacy Salesforce data pipelines to Databricks, utilizing FiveTran and JDBC to build scalable data ingestion workflows.
- Processed and integrated large datasets from multiple sources while maintaining high data quality and system reliability.

Python-Based Data Automation Tools | Python, Pandas, Selenium, Excel

- Developed automation scripts for data collection and processing by extracting parameters from PLM tools and Excel datasets.
- Streamlined complex data workflows, significantly improving process efficiency and standardizing output generation.

Unstructured Data Extraction & Processing | Python, PyPDF2, OpenCV, NumPy

- Built a customized Python and OCR-enabled tool to parse, extract, and clean data from digital PDFs.
- Leveraged automated workflows to reduce manual processing time by 6x, enhancing analytical capability and data accuracy.

Social Engagements

Sports Engagements: Chess, Cricket, Table-Tennis

Music Engagements: Former Guitarist at Riyaaz - Music Club of Kalyani Government Engineering College, and "Asokto - The Band"